

MATHEMATICAL STATISTICS FINAL REVIEW

CHAPTERS 1-9 COMPREHENSIVE GUIDE

STATISTICAL THEORY REVIEW

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1. PROBABILITY AND DISTRIBUTIONS (CHAPTER 1)

1.1. Fundamental Concepts.

Definition 1.1 (Probability Space). *A probability space is a triple $(\Omega, \mathcal{F}, P)$ where:*

- Ω is the sample space
- $\mathcal{F}$ is a σ -algebra on Ω
- $P : \mathcal{F} \rightarrow [0, 1]$ is a probability measure

Theorem 1.1 (Kolmogorov Axioms). *For any probability measure P :*

- (1) $P(\Omega) = 1$
- (2) $P(A) \geq 0$ for all $A \in \mathcal{F}$
- (3) For disjoint events $A_1, A_2, \dots$: $P(\bigcup_{i=1}^{\infty} A_i) = \sum_{i=1}^{\infty} P(A_i)$

1.2. Conditional Probability and Independence.

Definition 1.2 (Conditional Probability).

$$P(A|B) = \frac{P(A \cap B)}{P(B)}, \quad P(B) > 0$$

Theorem 1.2 (Bayes' Theorem). *Let $B_1, B_2, \dots, B_n$ be a partition of Ω . Then:*

$$P(B_i|A) = \frac{P(A|B_i)P(B_i)}{\sum_{j=1}^n P(A|B_j)P(B_j)}$$

Definition 1.3 (Independence). *Events A and B are independent if $P(A \cap B) = P(A)P(B)$. Random variables X and Y are independent if $F_{X,Y}(x, y) = F_X(x)F_Y(y)$.*

1.3. Random Variables and Distributions.

Definition 1.4 (Cumulative Distribution Function).

$$F_X(x) = P(X \leq x)$$

Properties:

- (1) Non-decreasing: $x_1 < x_2 \Rightarrow F_X(x_1) \leq F_X(x_2)$
- (2) Right-continuous: $\lim_{h \rightarrow 0^+} F_X(x + h) = F_X(x)$
- (3) $\lim_{x \rightarrow -\infty} F_X(x) = 0, \lim_{x \rightarrow \infty} F_X(x) = 1$

Definition 1.5 (Probability Mass/Density Functions). **Discrete:** $f_X(x) = P(X = x)$

Continuous: $f_X(x) = \frac{dF_X(x)}{dx}$ where $F_X(x) = \int_{-\infty}^x f_X(t)dt$

1.4. Expectation and Moments.

Definition 1.6 (Expectation). **Discrete:** $E[X] = \sum_x x f_X(x)$

Continuous: $E[X] = \int_{-\infty}^{\infty} x f_X(x)dx$

Theorem 1.3 (Properties of Expectation). (1) *Linearity:* $E[aX + bY] = aE[X] + bE[Y]$

- (2) *If $X \geq 0$, then $E[X] \geq 0$*
- (3) *If X and Y are independent: $E[XY] = E[X]E[Y]$*
- (4) *Law of Total Expectation: $E[X] = E[E[X|Y]]$*

Definition 1.7 (Variance).

$$\text{Var}(X) = E[(X - E[X])^2] = E[X^2] - (E[X])^2$$

Properties:

- $\text{Var}(aX + b) = a^2 \text{Var}(X)$
- *If X and Y are independent: $\text{Var}(X + Y) = \text{Var}(X) + \text{Var}(Y)$*
- *Law of Total Variance: $\text{Var}(X) = E[\text{Var}(X|Y)] + \text{Var}(E[X|Y])$*

1.5. Important Inequalities.

Theorem 1.4 (Markov's Inequality). *For any non-negative random variable X and $a > 0$:*

$$P(X \geq a) \leq \frac{E[X]}{a}$$

Theorem 1.5 (Chebyshev's Inequality). *For any random variable X with finite mean μ and variance σ^2 , and any $k > 0$:*

$$P(|X - \mu| \geq k\sigma) \leq \frac{1}{k^2}$$

Theorem 1.6 (Jensen's Inequality). *If g is a convex function and $E[X]$ exists, then:*

$$g(E[X]) \leq E[g(X)]$$

If g is concave, the inequality is reversed.

Theorem 1.7 (Cauchy-Schwarz Inequality). *For random variables X and Y with finite second moments:*

$$|E[XY]| \leq \sqrt{E[X^2]E[Y^2]}$$

Equality holds if and only if X and Y are linearly dependent.

2. MULTIVARIATE DISTRIBUTIONS (CHAPTER 2)

2.1. Joint Distributions.

Definition 2.1 (Joint CDF).

$$F_{X,Y}(x, y) = P(X \leq x, Y \leq y)$$

Definition 2.2 (Marginal Distributions).

$$F_X(x) = F_{X,Y}(x, \infty) = \lim_{y \rightarrow \infty} F_{X,Y}(x, y)$$

For continuous case:

$$f_X(x) = \int_{-\infty}^{\infty} f_{X,Y}(x, y) dy$$

2.2. Conditional Distributions.

Definition 2.3 (Conditional PDF).

$$f_{Y|X}(y|x) = \frac{f_{X,Y}(x, y)}{f_X(x)}, \quad f_X(x) > 0$$

Definition 2.4 (Conditional Expectation).

$$E[Y|X = x] = \int_{-\infty}^{\infty} y f_{Y|X}(y|x) dy$$

Law of Total Expectation: $E[Y] = E[E[Y|X]]$

2.3. Covariance and Correlation.

Definition 2.5 (Covariance).

$$\text{Cov}(X, Y) = E[(X - E[X])(Y - E[Y])] = E[XY] - E[X]E[Y]$$

Definition 2.6 (Correlation Coefficient).

$$\rho(X, Y) = \frac{\text{Cov}(X, Y)}{\sqrt{\text{Var}(X)\text{Var}(Y)}}$$

Properties: $-1 \leq \rho(X, Y) \leq 1$

2.4. Transformations.

Theorem 2.1 (Jacobian Method). *If $(U, V) = g(X, Y)$ where g is bijective with Jacobian J , then:*

$$f_{U,V}(u, v) = f_{X,Y}(g^{-1}(u, v))|J|^{-1}$$

$$\text{where } J = \det \begin{pmatrix} \frac{\partial x}{\partial u} & \frac{\partial x}{\partial v} \\ \frac{\partial y}{\partial u} & \frac{\partial y}{\partial v} \end{pmatrix}$$

3. SPECIAL DISTRIBUTIONS (CHAPTER 3)

3.1. Discrete Distributions.

Definition 3.1 (Binomial Distribution). $X \sim \text{Binomial}(n, p)$:

$$P(X = k) = \binom{n}{k} p^k (1-p)^{n-k}, \quad k = 0, 1, \dots, n$$

$$E[X] = np, \quad \text{Var}(X) = np(1-p)$$

Definition 3.2 (Poisson Distribution). $X \sim \text{Poisson}(\lambda)$:

$$P(X = k) = \frac{\lambda^k e^{-\lambda}}{k!}, \quad k = 0, 1, 2, \dots$$

$$E[X] = \text{Var}(X) = \lambda$$

Definition 3.3 (Negative Binomial Distribution). $X \sim \text{NegBin}(r, p)$ (number of failures before r -th success):

$$P(X = k) = \binom{k+r-1}{k} p^r (1-p)^k$$

$$E[X] = \frac{r(1-p)}{p}, \quad \text{Var}(X) = \frac{r(1-p)}{p^2}$$

3.2. Continuous Distributions.

Definition 3.4 (Normal Distribution). $X \sim \mathcal{N}(\mu, \sigma^2)$:

$$f_X(x) = \frac{1}{\sqrt{2\pi}\sigma} \exp\left(-\frac{(x-\mu)^2}{2\sigma^2}\right)$$

$$E[X] = \mu, \quad \text{Var}(X) = \sigma^2$$

Definition 3.5 (Gamma Distribution). $X \sim \text{Gamma}(\alpha, \beta)$:

$$f_X(x) = \frac{\beta^\alpha}{\Gamma(\alpha)} x^{\alpha-1} e^{-\beta x}, \quad x > 0$$

$$E[X] = \frac{\alpha}{\beta}, \quad \text{Var}(X) = \frac{\alpha}{\beta^2}$$

Definition 3.6 (Chi-Square Distribution). $X \sim \chi^2(\nu)$ where ν is degrees of freedom:

$$f_X(x) = \frac{1}{2^{\nu/2} \Gamma(\nu/2)} x^{\nu/2-1} e^{-x/2}, \quad x > 0$$

$$E[X] = \nu, \quad \text{Var}(X) = 2\nu$$

Note: $\chi^2(\nu) = \text{Gamma}(\nu/2, 1/2)$

Definition 3.7 (Student's t-Distribution). $T \sim t(\nu)$ where $T = \frac{Z}{\sqrt{V/\nu}}$, $Z \sim \mathcal{N}(0, 1)$, $V \sim \chi^2(\nu)$:

$$f_T(t) = \frac{\Gamma((\nu+1)/2)}{\sqrt{\nu\pi}\Gamma(\nu/2)} \left(1 + \frac{t^2}{\nu}\right)^{-(\nu+1)/2}$$

$$E[T] = 0 \text{ (if } \nu > 1), \quad \text{Var}(T) = \frac{\nu}{\nu-2} \text{ (if } \nu > 2)$$

Definition 3.8 (F-Distribution). $F \sim F(m, n)$ where $F = \frac{U/m}{V/n}$, $U \sim \chi^2(m)$, $V \sim \chi^2(n)$:

$$E[F] = \frac{n}{n-2} \text{ (if } n > 2)$$

Definition 3.9 (Uniform Distribution). $X \sim \text{Uniform}(a, b)$:

$$f_X(x) = \frac{1}{b-a}, \quad a < x < b$$

$$E[X] = \frac{a+b}{2}, \quad \text{Var}(X) = \frac{(b-a)^2}{12}$$

Definition 3.10 (Exponential Distribution). $X \sim \text{Exponential}(\lambda)$:

$$f_X(x) = \lambda e^{-\lambda x}, \quad x > 0$$

$$E[X] = \frac{1}{\lambda}, \quad \text{Var}(X) = \frac{1}{\lambda^2}$$

Memoryless Property: $P(X > s+t | X > s) = P(X > t)$

Definition 3.11 (Beta Distribution). $X \sim \text{Beta}(\alpha, \beta)$:

$$f_X(x) = \frac{\Gamma(\alpha+\beta)}{\Gamma(\alpha)\Gamma(\beta)} x^{\alpha-1} (1-x)^{\beta-1}, \quad 0 < x < 1$$

$$E[X] = \frac{\alpha}{\alpha+\beta}, \quad \text{Var}(X) = \frac{\alpha\beta}{(\alpha+\beta)^2(\alpha+\beta+1)}$$

Definition 3.12 (Hypergeometric Distribution). $X \sim \text{Hypergeometric}(N, K, n)$ (sampling without replacement):

$$P(X = k) = \frac{\binom{K}{k} \binom{N-K}{n-k}}{\binom{N}{n}}$$

$$E[X] = n \frac{K}{N}, \quad \text{Var}(X) = n \frac{K}{N} \frac{N-K}{N} \frac{N-n}{N-1}$$

3.3. Moment Generating Functions.

Definition 3.13 (Moment Generating Function). For random variable X :

$$M_X(t) = E[e^{tX}] = \begin{cases} \sum_x e^{tx} P(X=x) & \text{discrete} \\ \int_{-\infty}^{\infty} e^{tx} f_X(x) dx & \text{continuous} \end{cases}$$

Theorem 3.1 (Properties of MGF). (1) $M_X^{(n)}(0) = E[X^n]$ (moments)
 (2) If $Y = aX + b$, then $M_Y(t) = e^{bt} M_X(at)$
 (3) If X and Y are independent, then $M_{X+Y}(t) = M_X(t) M_Y(t)$
 (4) MGF uniquely determines the distribution

Example 3.1 (Common MGFs). • Normal: $M_X(t) = \exp(\mu t + \frac{\sigma^2 t^2}{2})$
 • Binomial: $M_X(t) = (1-p + pe^t)^n$
 • Poisson: $M_X(t) = \exp(\lambda(e^t - 1))$

- *Exponential*: $M_X(t) = \frac{\lambda}{\lambda - t}$ for $t < \lambda$
- *Gamma*: $M_X(t) = \left(\frac{\beta}{\beta - t}\right)^\alpha$ for $t < \beta$
- *Chi-square*: $M_X(t) = (1 - 2t)^{-k/2}$ for $t < \frac{1}{2}$

Definition 3.14 (Characteristic Function). $\phi_X(t) = E[e^{itX}]$ where $i = \sqrt{-1}$.

Properties:

- Always exists for any random variable
- $|\phi_X(t)| \leq 1$ for all t , $\phi_X(0) = 1$
- If X and Y independent: $\phi_{X+Y}(t) = \phi_X(t)\phi_Y(t)$
- Uniquely determines the distribution

3.4. Multivariate Normal Distribution.

Definition 3.15 (Bivariate Normal). $(X, Y) \sim \mathcal{N}_2(\boldsymbol{\mu}, \boldsymbol{\Sigma})$ where $\boldsymbol{\mu} = (\mu_1, \mu_2)^T$ and

$$\boldsymbol{\Sigma} = \begin{pmatrix} \sigma_1^2 & \rho\sigma_1\sigma_2 \\ \rho\sigma_1\sigma_2 & \sigma_2^2 \end{pmatrix}$$

Joint pdf:

$$f(x, y) = \frac{1}{2\pi\sigma_1\sigma_2\sqrt{1-\rho^2}} \exp\left(-\frac{Q}{2(1-\rho^2)}\right)$$

where $Q = \frac{(x-\mu_1)^2}{\sigma_1^2} - \frac{2\rho(x-\mu_1)(y-\mu_2)}{\sigma_1\sigma_2} + \frac{(y-\mu_2)^2}{\sigma_2^2}$

4. ELEMENTARY STATISTICAL INFERENCES (CHAPTER 4)

4.1. Sampling and Statistics.

Definition 4.1 (Random Sample). $X_1, X_2, \dots, X_n$ is a random sample if they are independent and identically distributed (i.i.d.).

Definition 4.2 (Sample Statistics). • *Sample mean*: $\bar{X} = \frac{1}{n} \sum_{i=1}^n X_i$

- *Sample variance*: $S^2 = \frac{1}{n-1} \sum_{i=1}^n (X_i - \bar{X})^2$
- *Sample standard deviation*: $S = \sqrt{S^2}$

Theorem 4.1 (Sampling from Normal Distribution). If $X_1, \dots, X_n \stackrel{iid}{\sim} \mathcal{N}(\mu, \sigma^2)$, then:

- (1) $\bar{X} \sim \mathcal{N}(\mu, \sigma^2/n)$
- (2) $\frac{(n-1)S^2}{\sigma^2} \sim \chi^2(n-1)$
- (3) $\bar{X}$ and S^2 are independent
- (4) $\frac{\bar{X}-\mu}{S/\sqrt{n}} \sim t(n-1)$

4.2. Confidence Intervals.

Definition 4.3 (Confidence Interval). A $100(1 - \alpha)\%$ confidence interval for parameter θ is an interval $[L, U]$ such that:

$$P(L \leq \theta \leq U) = 1 - \alpha$$

Theorem 4.2 (Confidence Intervals for Normal Mean). **Case 1: σ^2 known**

$$\bar{X} \pm z_{\alpha/2} \frac{\sigma}{\sqrt{n}}$$

Case 2: σ^2 unknown

$$\bar{X} \pm t_{\alpha/2}(n-1) \frac{S}{\sqrt{n}}$$

4.3. Order Statistics.

Definition 4.4 (Order Statistics). For sample $X_1, \dots, X_n$, the order statistics are $X_{(1)} \leq X_{(2)} \leq \dots \leq X_{(n)}$.

The k -th order statistic $X_{(k)}$ has pdf:

$$f_{X_{(k)}}(x) = \frac{n!}{(k-1)!(n-k)!} [F(x)]^{k-1} [1 - F(x)]^{n-k} f(x)$$

4.4. Introduction to Hypothesis Testing.

Definition 4.5 (Hypothesis Test). • Null hypothesis: H_0

- Alternative hypothesis: H_1
- Test statistic: $T(\mathbf{X})$
- Critical region: $\mathcal{C}$
- Decision rule: Reject H_0 if $T(\mathbf{X}) \in \mathcal{C}$

Definition 4.6 (Type I and Type II Errors). • Type I error: Reject H_0 when H_0 is true, $\alpha = P(\text{Type I error})$

- Type II error: Accept H_0 when H_1 is true, $\beta = P(\text{Type II error})$
- Power: $1 - \beta = P(\text{Reject } H_0 | H_1 \text{ true})$

4.5. Wald Test.

Definition 4.7 (Wald Test). For testing $H_0 : \theta = \theta_0$ vs $H_1 : \theta \neq \theta_0$, when $\hat{\theta}$ is asymptotically normal:

$$W = \frac{\hat{\theta} - \theta_0}{\widehat{se}(\hat{\theta})} \xrightarrow{d} N(0, 1)$$

Reject H_0 if $|W| > z_{\alpha/2}$.

4.6. p-values.

Definition 4.8 (p-value). The p-value is the probability, under H_0 , of observing a test statistic as extreme or more extreme than what was actually observed.

For two-sided test: $p\text{-value} = 2P(|T| \geq |t_{\text{obs}}| | H_0)$

4.7. Chi-Square Tests.

Definition 4.9 (Goodness-of-Fit Test). For testing if data follows a specified distribution:

$$\chi^2 = \sum_{i=1}^k \frac{(O_i - E_i)^2}{E_i} \sim \chi^2(k - 1 - r)$$

where O_i = observed frequency, E_i = expected frequency, r = number of estimated parameters.

Definition 4.10 (Test of Independence). For testing independence in contingency tables:

$$\chi^2 = \sum_{i=1}^r \sum_{j=1}^c \frac{(O_{ij} - E_{ij})^2}{E_{ij}} \sim \chi^2((r-1)(c-1))$$

where $E_{ij} = \frac{n_{i.}n_{.j}}{n}$.

5. CONSISTENCY AND LIMITING DISTRIBUTIONS (CHAPTER 5)

5.1. Modes of Convergence.

Definition 5.1 (Convergence in Probability). $X_n \xrightarrow{p} X$ if for any $\epsilon > 0$:

$$\lim_{n \rightarrow \infty} P(|X_n - X| > \epsilon) = 0$$

Definition 5.2 (Convergence in Distribution). $X_n \xrightarrow{d} X$ if $F_n(x) \rightarrow F(x)$ at all continuity points of F .

Theorem 5.1 (Relationships Between Convergence Types).

$$a.s. \Rightarrow probability \Rightarrow distribution$$

$$L^p \Rightarrow probability$$

5.2. Law of Large Numbers.

Theorem 5.2 (Weak Law of Large Numbers). If $X_1, X_2, \dots$ are i.i.d. with $E[X_i] = \mu$ and $Var(X_i) = \sigma^2 < \infty$, then:

$$\bar{X}_n \xrightarrow{p} \mu$$

Theorem 5.3 (Strong Law of Large Numbers). If $X_1, X_2, \dots$ are i.i.d. with $E[|X_i|] < \infty$ and $E[X_i] = \mu$, then:

$$\bar{X}_n \xrightarrow{a.s.} \mu$$

5.3. Central Limit Theorem.

Theorem 5.4 (Central Limit Theorem). If $X_1, X_2, \dots$ are i.i.d. with $E[X_i] = \mu$ and $Var(X_i) = \sigma^2 < \infty$, then:

$$\frac{\sqrt{n}(\bar{X}_n - \mu)}{\sigma} \xrightarrow{d} \mathcal{N}(0, 1)$$

5.4. Delta Method.

Theorem 5.5 (Delta Method). If $\sqrt{n}(X_n - \theta) \xrightarrow{d} \mathcal{N}(0, \sigma^2)$ and g is differentiable at θ with $g'(\theta) \neq 0$, then:

$$\sqrt{n}(g(X_n) - g(\theta)) \xrightarrow{d} \mathcal{N}(0, [g'(\theta)]^2 \sigma^2)$$

6. MAXIMUM LIKELIHOOD METHODS (CHAPTER 6)

6.1. Maximum Likelihood Estimation.

Definition 6.1 (Likelihood Function). For sample $\mathbf{x} = (x_1, \dots, x_n)$:

$$L(\theta; \mathbf{x}) = \prod_{i=1}^n f(x_i; \theta)$$

$$\text{Log-likelihood: } \ell(\theta; \mathbf{x}) = \log L(\theta; \mathbf{x}) = \sum_{i=1}^n \log f(x_i; \theta)$$

Definition 6.2 (Maximum Likelihood Estimator).

$$\hat{\theta}_{MLE} = \arg \max_{\theta} L(\theta; \mathbf{x})$$

Often found by solving: $\frac{\partial \ell(\theta)}{\partial \theta} = 0$

6.2. Fisher Information.

Definition 6.3 (Fisher Information).

$$I(\theta) = -E \left[\frac{\partial^2 \log f(X; \theta)}{\partial \theta^2} \right] = E \left[\left(\frac{\partial \log f(X; \theta)}{\partial \theta} \right)^2 \right]$$

For sample of size n : $I_n(\theta) = nI(\theta)$

Theorem 6.1 (Cramér-Rao Lower Bound). For any unbiased estimator $\hat{\theta}$:

$$\text{Var}(\hat{\theta}) \geq \frac{1}{I_n(\theta)}$$

6.3. Asymptotic Properties of MLE.

Theorem 6.2 (Asymptotic Properties of MLE). Under regularity conditions:

- (1) Consistency: $\hat{\theta}_n \xrightarrow{p} \theta_0$
- (2) Asymptotic normality: $\sqrt{n}(\hat{\theta}_n - \theta_0) \xrightarrow{d} \mathcal{N}(0, I^{-1}(\theta_0))$
- (3) Asymptotic efficiency: Achieves Cramér-Rao bound asymptotically

6.4. Likelihood Ratio Tests.

Definition 6.4 (Likelihood Ratio Test Statistic). For testing $H_0 : \theta \in \Theta_0$ vs $H_1 : \theta \in \Theta_1$:

$$\Lambda(\mathbf{x}) = \frac{\sup_{\theta \in \Theta_0} L(\theta; \mathbf{x})}{\sup_{\theta \in \Theta} L(\theta; \mathbf{x})}$$

Theorem 6.3 (Wilks' Theorem). Under H_0 and regularity conditions:

$$-2 \log \Lambda(\mathbf{X}) \xrightarrow{d} \chi^2(r)$$

where $r = \dim(\Theta) - \dim(\Theta_0)$.

7. SUFFICIENCY (CHAPTER 7)

7.1. Sufficient Statistics.

Definition 7.1 (Sufficient Statistic). $T(\mathbf{X})$ is sufficient for θ if the conditional distribution of $\mathbf{X}$ given $T(\mathbf{X})$ does not depend on θ .

Theorem 7.1 (Factorization Theorem). $T(\mathbf{X})$ is sufficient for θ if and only if:

$$f(\mathbf{x}; \theta) = g(T(\mathbf{x}); \theta) \cdot h(\mathbf{x})$$

where g depends on $\mathbf{x}$ only through $T(\mathbf{x})$ and h doesn't depend on θ .

7.2. Minimal Sufficiency and Completeness.

Definition 7.2 (Minimal Sufficient Statistic). T is minimal sufficient if for any other sufficient statistic T' , there exists a function f such that $T = f(T')$.

Definition 7.3 (Complete Statistic). T is complete if $E[g(T)] = 0$ for all θ implies $P(g(T) = 0) = 1$ for all θ .

Theorem 7.2 (Rao-Blackwell Theorem). If $\hat{\theta}$ is unbiased for θ and T is sufficient, then $\hat{\theta}^* = E[\hat{\theta}|T]$ satisfies:

- (1) $E[\hat{\theta}^*] = \theta$ (unbiased)
- (2) $\text{Var}(\hat{\theta}^*) \leq \text{Var}(\hat{\theta})$

Theorem 7.3 (Lehmann-Scheffé Theorem). If T is complete and sufficient, then there exists a unique UMVUE based on T .

7.3. Methods for Determining Sufficiency and Completeness.

Theorem 7.4 (Neyman Factorization Theorem (Detailed)). $Y_1 = u_1(X_1, \dots, X_n)$ is sufficient for θ if and only if the joint pdf/pmf can be factored as:

$$f(x_1; \theta) \cdots f(x_n; \theta) = k_1[u_1(x_1, \dots, x_n); \theta] \times k_2(x_1, \dots, x_n)$$

where k_2 does not depend on θ , and k_1 depends on the data only through Y_1 .

Definition 7.4 (Complete Family of Distributions). A family of pdfs/pmfs $\{h(z; \theta) : \theta \in \Omega\}$ for a statistic Z is **complete** if:

$$E[u(Z)] = 0 \text{ for all } \theta \in \Omega \Rightarrow u(z) = 0 \text{ (almost everywhere)}$$

Theorem 7.5 (Exponential Family and Complete Sufficiency). A family $\{f(x; \theta) : \theta \in \Omega\}$ is a **regular exponential class** if:

$$f(x; \theta) = \exp[p(\theta)K(x) + H(x) + q(\theta)]$$

where:

- (1) The support does not depend on θ
- (2) $p(\theta)$ is a nontrivial continuous function of θ
- (3) $K(x)$ is nontrivial and $H(x)$ is continuous

For a random sample from such a distribution, $Y_1 = \sum_{i=1}^n K(X_i)$ is a **complete sufficient statistic**.

Example 7.1 (Normal Distribution (σ^2 known)). For $X_1, \dots, X_n \sim N(\theta, \sigma^2)$ with σ^2 known:

$$f(x; \theta) = \exp \left[\frac{\theta}{\sigma^2} x - \frac{x^2}{2\sigma^2} - \frac{\theta^2}{2\sigma^2} - \frac{1}{2} \log(2\pi\sigma^2) \right]$$

Here $K(x) = x$, so $Y_1 = \sum X_i = n\bar{X}$ is complete sufficient for θ . Since $E[\bar{X}] = \theta$, $\bar{X}$ is the unique MVUE of θ .

Example 7.2 (Poisson Distribution). For $X_1, \dots, X_n \sim \text{Poisson}(\theta)$:

$$f(x; \theta) = \exp[x \log \theta - \theta - \log(x!)]$$

Here $K(x) = x$, so $Y_1 = \sum X_i$ is complete sufficient for θ . Since $E[Y_1/n] = \theta$, $\bar{X} = Y_1/n$ is the unique MVUE of θ .

Example 7.3 (Uniform Distribution). For $X_1, \dots, X_n \sim \text{Uniform}(0, \theta)$: - $Y_n = \max(X_1, \dots, X_n)$ is sufficient for θ - The pdf of Y_n is $g(y_n; \theta) = \frac{ny_n^{n-1}}{\theta^n}$ for $0 < y_n < \theta$ - This family is complete - Since $E[Y_n] = \frac{n}{n+1}\theta$, the MVUE of θ is $\frac{n+1}{n}Y_n$

7.4. Minimal Sufficiency.

Definition 7.5 (Minimal Sufficient Statistic). A sufficient statistic T is **minimal sufficient** if it is a function of every other sufficient statistic. Equivalently, T achieves the maximum possible data reduction while retaining sufficiency.

Theorem 7.6 (Lehmann-Scheffé Criterion for Minimal Sufficiency). A statistic $T(\mathbf{X})$ is minimal sufficient if for any two data points $\mathbf{x}$ and $\mathbf{z}$:

$$\frac{L(\theta; \mathbf{x})}{L(\theta; \mathbf{z})} \text{ is independent of } \theta \Leftrightarrow T(\mathbf{x}) = T(\mathbf{z})$$

Remark 7.1. complete sufficient statistic is minimal sufficient. but minimal sufficient is not necessarily complete sufficient.

7.5. Ancillary Statistics and Basu's Theorem.

Definition 7.6 (Ancillary Statistic). A statistic Z is **ancillary** for θ if its distribution does not depend on θ .

Common Types:

- **Location-invariant:** For $X_i = \theta + W_i$, statistics like S^2 and Range
- **Scale-invariant:** For $X_i = \theta W_i$, statistics like $X_1/(X_1 + X_2)$
- **Location and scale-invariant:** Statistics like $(X_i - \bar{X})/S$

Theorem 7.7 (Basu's Theorem). If Y_1 is a **complete sufficient statistic** for θ , then Y_1 is independent of any **ancillary statistic** Z .

Converse: If Y_1 is sufficient and independent of an ancillary statistic Z , then Y_1 is complete.

Example 7.4 (Independence of $\bar{X}$ and S^2). For $X_1, \dots, X_n \sim N(\mu, \sigma^2)$:

- (1) When σ^2 is known: $\bar{X}$ is complete sufficient for μ , S^2 is ancillary for μ
- (2) By Basu's theorem: $\bar{X}$ and S^2 are independent

7.6. Multi-Parameter Cases.

Theorem 7.8 (Joint Sufficiency). A vector of statistics $\mathbf{Y} = (Y_1, \dots, Y_m)$ is **jointly sufficient** for $\boldsymbol{\theta} = (\theta_1, \dots, \theta_p)$ if:

$$\prod_{i=1}^n f(x_i; \boldsymbol{\theta}) = k_1(\mathbf{y}; \boldsymbol{\theta}) \times k_2(x_1, \dots, x_n)$$

where k_2 does not depend on $\boldsymbol{\theta}$.

Theorem 7.9 (Multiparameter Exponential Family). For the multiparameter exponential family:

$$f(x; \boldsymbol{\theta}) = \exp \left[\sum_{j=1}^m p_j(\boldsymbol{\theta}) K_j(x) + H(x) + q(\boldsymbol{\theta}) \right]$$

The statistics $Y_j = \sum_{i=1}^n K_j(X_i)$ for $j = 1, \dots, m$ are **jointly complete sufficient**.

Example 7.5 (Normal Distribution (Both Parameters Unknown)). For $X_1, \dots, X_n \sim N(\mu, \sigma^2)$ with both parameters unknown:

$$f(x; \mu, \sigma^2) = \exp \left[\frac{\mu}{\sigma^2} x - \frac{1}{2\sigma^2} x^2 - \frac{\mu^2}{2\sigma^2} - \frac{1}{2} \log(2\pi\sigma^2) \right]$$

- (1) $K_1(x) = x$, $K_2(x) = x^2$
- (2) $Y_1 = \sum X_i$, $Y_2 = \sum X_i^2$ are jointly complete sufficient
- (3) Equivalently: $\bar{X}$ and S^2 are jointly complete sufficient
- (4) Since $E[\bar{X}] = \mu$ and $E[S^2] = \sigma^2$, they are the unique MVUEs

7.7. Estimating Functions of Parameters.

Theorem 7.10 (MVUE of $g(\theta)$). If Y_1 is complete sufficient for θ , and we want to estimate $g(\theta)$:

- (1) Find any unbiased estimator T of $g(\theta)$ (not necessarily a function of Y_1)
- (2) Compute $\psi(Y_1) = E[T|Y_1]$ (Rao-Blackwell improvement)
- (3) Then $\psi(Y_1)$ is the unique MVUE of $g(\theta)$

Alternatively, if you can find a function $\psi(Y_1)$ such that $E[\psi(Y_1)] = g(\theta)$, then $\psi(Y_1)$ is automatically the unique MVUE.

Example 7.6 (Bernoulli Variance). For $X_1, \dots, X_n \sim \text{Bernoulli}(\theta)$, to estimate $\delta = \theta(1 - \theta)$:

- (1) $Y = \sum X_i$ is complete sufficient for θ
- (2) MLE of δ is $\hat{\delta} = \bar{X}(1 - \bar{X}) = \frac{Y}{n}(1 - \frac{Y}{n})$
- (3) $E[\hat{\delta}] = \frac{n-1}{n}\theta(1 - \theta) = \frac{n-1}{n}\delta$ (biased)
- (4) MVUE of δ is $\frac{n}{n-1}\hat{\delta} = \frac{n}{n-1}\frac{Y}{n}(1 - \frac{Y}{n})$

Example 7.7 (Exponential Distribution). For $X_1, \dots, X_n \sim \text{Exponential}(\theta)$ with mean $1/\theta$:

- (1) $Y_1 = \sum X_i$ is complete sufficient
- (2) MLE: $\hat{\theta} = n/Y_1$ (biased, $E[\hat{\theta}] = \frac{n\theta}{n-1}$)
- (3) MVUE: $\frac{n-1}{Y_1}$ (since $E[\frac{n-1}{Y_1}] = \theta$)

8. OPTIMAL TESTS OF HYPOTHESES (CHAPTER 8)

8.1. Neyman-Pearson Theory.

Theorem 8.1 (Neyman-Pearson Lemma). For testing simple $H_0 : \theta = \theta_0$ vs simple $H_1 : \theta = \theta_1$, the most powerful test of size α rejects H_0 when:

$$\frac{L(\theta_1; \mathbf{x})}{L(\theta_0; \mathbf{x})} > k$$

where k is chosen so that $P_{\theta_0}(\text{reject } H_0) = \alpha$.

Definition 8.1 (Uniformly Most Powerful Test). A test is UMP of size α for $H_0 : \theta \in \Theta_0$ vs $H_1 : \theta \in \Theta_1$ if:

- (1) It has size α
- (2) Among all tests of size $\leq \alpha$, it has maximum power for every $\theta \in \Theta_1$

8.2. Likelihood Ratio Tests.

Theorem 8.2 (Generalized Likelihood Ratio Test). For composite hypotheses, reject H_0 when $\Lambda(\mathbf{x}) \leq c$ where:

$$\Lambda(\mathbf{x}) = \frac{\sup_{\theta \in \Theta_0} L(\theta; \mathbf{x})}{\sup_{\theta \in \Theta} L(\theta; \mathbf{x})}$$

9. NORMAL LINEAR MODELS (CHAPTER 9)

9.1. One-Way ANOVA.

Definition 9.1 (One-Way ANOVA Model).

$$X_{ij} = \mu + \alpha_i + \epsilon_{ij}$$

where $i = 1, \dots, k$, $j = 1, \dots, n_i$, and $\epsilon_{ij} \stackrel{iid}{\sim} \mathcal{N}(0, \sigma^2)$.

Constraint: $\sum_{i=1}^k n_i \alpha_i = 0$

Theorem 9.1 (ANOVA F-Test). Test $H_0 : \alpha_1 = \dots = \alpha_k = 0$ vs H_1 : not all $\alpha_i = 0$.

Test statistic: $F = \frac{MSA}{MSE} = \frac{SSA/(k-1)}{SSE/(N-k)}$

Under H_0 : $F \sim F(k-1, N-k)$ where $N = \sum_{i=1}^k n_i$.

Theorem 9.2 (Sum of Squares Decomposition).

$$SST = SSA + SSE$$

where:

- $SST = \sum_{i=1}^k \sum_{j=1}^{n_i} (X_{ij} - \bar{X}_{..})^2$ (Total Sum of Squares)
- $SSA = \sum_{i=1}^k n_i (\bar{X}_{i.} - \bar{X}_{..})^2$ (Treatment Sum of Squares)
- $SSE = \sum_{i=1}^k \sum_{j=1}^{n_i} (X_{ij} - \bar{X}_{i.})^2$ (Error Sum of Squares)

9.2. Multiple Comparisons.

Definition 9.2 (Tukey's HSD (Honestly Significant Difference)). For equal sample sizes m :

$$HSD = q_\alpha(k, f_e) \sqrt{\frac{MSE}{m}}$$

Two means differ significantly if $|\bar{X}_{i.} - \bar{X}_{j.}| \geq HSD$.

Definition 9.3 (Bonferroni Correction). For c comparisons at overall level α , use individual level α/c for each comparison.

Definition 9.4 (Scheffé Method). For any contrast $L = \sum c_i \mu_i$ where $\sum c_i = 0$:

$$S = \sqrt{(k-1)F_\alpha(k-1, f_e)} \sqrt{MSE \sum_{i=1}^k \frac{c_i^2}{n_i}}$$

9.3. Two-Way ANOVA.

Definition 9.5 (Two-Way ANOVA Model).

$$X_{ijk} = \mu + \alpha_i + \beta_j + (\alpha\beta)_{ij} + \epsilon_{ijk}$$

where:

- α_i : effect of factor A at level i
- β_j : effect of factor B at level j
- $(\alpha\beta)_{ij}$: interaction effect
- $\epsilon_{ijk} \sim N(0, \sigma^2)$

Constraints: $\sum \alpha_i = \sum \beta_j = \sum_i (\alpha\beta)_{ij} = \sum_j (\alpha\beta)_{ij} = 0$

Theorem 9.3 (Two-Way ANOVA Decomposition).

$$SST = SSA + SSB + SSAB + SSE$$

where:

- $SSA = bn \sum_{i=1}^a (\bar{X}_{i..} - \bar{X}_{...})^2$ (Factor A)
- $SSB = an \sum_{j=1}^b (\bar{X}_{.j.} - \bar{X}_{...})^2$ (Factor B)
- $SSAB = n \sum_{i=1}^a \sum_{j=1}^b (\bar{X}_{ij.} - \bar{X}_{i..} - \bar{X}_{.j.} + \bar{X}_{...})^2$ (Interaction)
- $SSE = \sum_{i=1}^a \sum_{j=1}^b \sum_{k=1}^n (X_{ijk} - \bar{X}_{ij.})^2$ (Error)

Theorem 9.4 (Two-Way ANOVA F-Tests). • Factor A: $F_A = \frac{MSA}{MSE} \sim F(a-1, ab(n-1))$

• Factor B: $F_B = \frac{MSB}{MSE} \sim F(b-1, ab(n-1))$

• Interaction: $F_{AB} = \frac{MSAB}{MSE} \sim F((a-1)(b-1), ab(n-1))$

9.4. Tests for Homogeneity of Variances.

Definition 9.6 (Bartlett's Test). For testing $H_0: \sigma_1^2 = \sigma_2^2 = \dots = \sigma_k^2$:

$$B = \frac{(N-k) \ln(s_p^2) - \sum_{i=1}^k (n_i - 1) \ln(s_i^2)}{1 + \frac{1}{3(k-1)} [\sum_{i=1}^k \frac{1}{n_i - 1} - \frac{1}{N-k}]}$$

where $s_p^2 = \frac{\sum_{i=1}^k (n_i - 1) s_i^2}{N-k}$ and $N = \sum_{i=1}^k n_i$. Under H_0 : $B \sim \chi^2(k-1)$.

Definition 9.7 (Levene's Test (Robust Alternative)). *Replace observations with $Z_{ij} = |X_{ij} - \tilde{X}_i|$ where $\tilde{X}_i$ is the group median. Apply one-way ANOVA to the Z_{ij} values.*

9.5. Simple Linear Regression.

Definition 9.8 (Simple Linear Regression Model).

$$Y_i = \beta_0 + \beta_1 x_i + \epsilon_i$$

where $\epsilon_i \stackrel{iid}{\sim} \mathcal{N}(0, \sigma^2)$.

Theorem 9.5 (Least Squares Estimators).

$$\hat{\beta}_1 = \frac{\sum_{i=1}^n (x_i - \bar{x})(Y_i - \bar{Y})}{\sum_{i=1}^n (x_i - \bar{x})^2} = \frac{S_{xy}}{S_{xx}}$$

$$\hat{\beta}_0 = \bar{Y} - \hat{\beta}_1 \bar{x}$$

where $S_{xy} = \sum (x_i - \bar{x})(Y_i - \bar{Y})$ and $S_{xx} = \sum (x_i - \bar{x})^2$.

Theorem 9.6 (Properties of LS Estimators). (1) $\hat{\beta}_1 \sim \mathcal{N}\left(\beta_1, \frac{\sigma^2}{S_{xx}}\right)$

(2) $\hat{\beta}_0 \sim \mathcal{N}\left(\beta_0, \sigma^2 \left(\frac{1}{n} + \frac{\bar{x}^2}{S_{xx}}\right)\right)$

(3) $\frac{(n-2)S^2}{\sigma^2} \sim \chi^2(n-2)$ where $S^2 = \frac{SSE}{n-2}$

9.6. Regression Analysis Details.

Theorem 9.7 (Sum of Squares Decomposition in Regression).

$$SST = SSR + SSE$$

where:

- $SST = \sum_{i=1}^n (Y_i - \bar{Y})^2$ (Total Sum of Squares)
- $SSR = \sum_{i=1}^n (\hat{Y}_i - \bar{Y})^2$ (Regression Sum of Squares)
- $SSE = \sum_{i=1}^n (Y_i - \hat{Y}_i)^2$ (Error Sum of Squares)

Definition 9.9 (Coefficient of Determination).

$$R^2 = \frac{SSR}{SST} = 1 - \frac{SSE}{SST}$$

R^2 represents the proportion of variation in Y explained by the regression model.

Theorem 9.8 (F-Test for Regression). To test $H_0 : \beta_1 = 0$ vs $H_1 : \beta_1 \neq 0$:

$$F = \frac{MSR}{MSE} = \frac{SSR/1}{SSE/(n-2)} \sim F(1, n-2)$$

Theorem 9.9 (t-Test for Regression Coefficients).

$$t = \frac{\hat{\beta}_1}{\sqrt{S^2/S_{xx}}} \sim t(n-2)$$

$$t = \frac{\hat{\beta}_0}{\sqrt{S^2(1/n + \bar{x}^2/S_{xx})}} \sim t(n-2)$$

9.7. Prediction and Confidence Intervals.

Theorem 9.10 (Confidence Interval for Mean Response). *At $x = x_0$, a $100(1 - \alpha)\%$ confidence interval for $E[Y|x_0]$ is:*

$$\hat{Y}_0 \pm t_{\alpha/2, n-2} \sqrt{S^2 \left(\frac{1}{n} + \frac{(x_0 - \bar{x})^2}{S_{xx}} \right)}$$

Theorem 9.11 (Prediction Interval for New Observation). *At $x = x_0$, a $100(1 - \alpha)\%$ prediction interval for a new observation Y_0 is:*

$$\hat{Y}_0 \pm t_{\alpha/2, n-2} \sqrt{S^2 \left(1 + \frac{1}{n} + \frac{(x_0 - \bar{x})^2}{S_{xx}} \right)}$$

9.8. Correlation Analysis.

Definition 9.10 (Sample Correlation Coefficient).

$$r = \frac{\sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y})}{\sqrt{\sum_{i=1}^n (x_i - \bar{x})^2 \sum_{i=1}^n (y_i - \bar{y})^2}} = \frac{S_{xy}}{\sqrt{S_{xx}S_{yy}}}$$

Properties: $-1 \leq r \leq 1$, and $r^2 = R^2$ in simple linear regression.

Theorem 9.12 (Test for Correlation). *To test $H_0 : \rho = 0$ vs $H_1 : \rho \neq 0$:*

$$t = \frac{r\sqrt{n-2}}{\sqrt{1-r^2}} \sim t(n-2)$$

9.9. Nonlinear Regression. Common transformations to linearize relationships:

- Exponential: $y = ae^{bx} \Rightarrow \ln(y) = \ln(a) + bx$
- Power: $y = ax^b \Rightarrow \ln(y) = \ln(a) + b \ln(x)$
- Logarithmic: $y = a + b \ln(x)$ (already linear in parameters)
- Reciprocal: $y = \frac{1}{a+bx} \Rightarrow \frac{1}{y} = a + bx$

Concept	Formula
Sample mean distribution	$\bar{X} \sim \mathcal{N}(\mu, \sigma^2/n)$
t-statistic	$\frac{\bar{X} - \mu}{S/\sqrt{n}} \sim t(n-1)$
Chi-square statistic	$\frac{(n-1)S^2}{\sigma^2} \sim \chi^2(n-1)$
Cramér-Rao bound	$\text{Var}(\hat{\theta}) \geq \frac{1}{nI(\theta)}$
MLE asymptotic	$\sqrt{n}(\hat{\theta} - \theta) \xrightarrow{d} \mathcal{N}(0, I^{-1}(\theta))$
LRT statistic	$-2 \log \Lambda \xrightarrow{d} \chi^2(r)$
Delta method	$\sqrt{n}(g(\hat{\theta}) - g(\theta)) \xrightarrow{d} \mathcal{N}(0, [g'(\theta)]^2/I(\theta))$

9.10. Key Formulas and Results Summary.

9.11. Common Test Statistics Summary.

9.12. Important Constants and Critical Values. Standard Normal:

- $z_{0.025} = 1.96$, $z_{0.05} = 1.645$, $z_{0.01} = 2.326$

Common t-values: For large n , t -distribution approaches standard normal.

Chi-square: $\chi_{0.05}^2(1) = 3.84$, $\chi_{0.05}^2(2) = 5.99$

Test	Statistic	Distribution
One-sample t-test	$\frac{\bar{X} - \mu_0}{S/\sqrt{n}}$	$t(n - 1)$
Two-sample t-test	$\frac{\bar{X}_1 - \bar{X}_2}{S_p \sqrt{1/n_1 + 1/n_2}}$	$t(n_1 + n_2 - 2)$
F-test (variances)	$\frac{S_1^2}{S_2^2}$	$F(n_1 - 1, n_2 - 1)$
Chi-square goodness-of-fit	$\sum \frac{(O_i - E_i)^2}{E_i}$	$\chi^2(k - 1 - r)$
ANOVA F-test	$\frac{MSA}{MSE}$	$F(k - 1, N - k)$
Regression F-test	$\frac{MSR}{MSE}$	$F(1, n - 2)$
Correlation test	$\frac{r\sqrt{n-2}}{\sqrt{1-r^2}}$	$t(n - 2)$

9.13. Practical Guidelines.

(1) Choosing Tests:

- Use t-test when σ unknown and n small
- Use z-test when σ known or n large
- Use F-test for comparing variances or in ANOVA
- Use χ^2 -test for categorical data or goodness-of-fit

(2) Sample Size Considerations:

- CLT applies for $n \geq 30$ (rule of thumb)
- For t-tests, small sample assumptions require normality
- ANOVA requires equal variances (Bartlett's or Levene's test)

(3) Model Assumptions:

- Independence: Crucial for all procedures
- Normality: Required for exact results, less critical for large n
- Equal variances: Important for pooled procedures
- Linearity: Essential for regression models

9.14. Final Exam Quick Reference. Must Remember Formulas:

- Sample variance: $S^2 = \frac{1}{n-1} \sum_{i=1}^n (X_i - \bar{X})^2$
- Standard error of mean: $SE(\bar{X}) = \frac{S}{\sqrt{n}}$
- Confidence interval: $\bar{X} \pm t_{\alpha/2, n-1} \frac{S}{\sqrt{n}}$
- Pooled variance: $S_p^2 = \frac{(n_1-1)S_1^2 + (n_2-1)S_2^2}{n_1 + n_2 - 2}$
- Regression slope: $\hat{\beta}_1 = \frac{\sum (x_i - \bar{x})(y_i - \bar{y})}{\sum (x_i - \bar{x})^2}$
- Coefficient of determination: $R^2 = \frac{SSR}{SST} = 1 - \frac{SSE}{SST}$

Key Distributions for Testing:

- t-distribution: For means when σ unknown
- F-distribution: For variances and ANOVA
- χ^2 -distribution: For variances and goodness-of-fit
- Normal: For large samples and known σ

Common Mistakes to Avoid:

- Confusing population and sample parameters (μ vs $\bar{x}$, σ^2 vs s^2)
- Using wrong degrees of freedom
- Forgetting to check assumptions (normality, independence, equal variances)
- Misinterpreting p-values and confidence intervals
- Not distinguishing between one-tailed and two-tailed tests

Distribution	PDF/PMF	Mean	Variance
Bernoulli(p)	$p^x(1-p)^{1-x}$	p	$p(1-p)$
Binomial(n, p)	$\binom{n}{x}p^x(1-p)^{n-x}$	np	$np(1-p)$
Poisson(λ)	$\frac{\lambda^x e^{-\lambda}}{x!}$	λ	λ
Uniform(a, b)	$\frac{1}{b-a}$	$\frac{a+b}{2}$	$\frac{(b-a)^2}{12}$
Normal(μ, σ^2)	$\frac{1}{\sqrt{2\pi\sigma^2}} e^{-\frac{(x-\mu)^2}{2\sigma^2}}$	μ	σ^2
Exponential(λ)	$\lambda e^{-\lambda x}$	$\frac{1}{\lambda}$	$\frac{1}{\lambda^2}$
Gamma(α, β)	$\frac{\beta^\alpha}{\Gamma(\alpha)} x^{\alpha-1} e^{-\beta x}$	$\frac{\alpha}{\beta}$	$\frac{\alpha}{\beta^2}$
Chi-square(ν)	$\frac{1}{2^{\nu/2}\Gamma(\nu/2)} x^{\nu/2-1} e^{-x/2}$	ν	2ν
t(ν)	$\frac{\Gamma((\nu+1)/2)}{\sqrt{\nu\pi}\Gamma(\nu/2)} (1 + \frac{t^2}{\nu})^{-(\nu+1)/2}$	0	$\frac{\nu}{\nu-2}$

A.1. A.1 Common Distributions Summary.

Distribution	MGF
Binomial(n, p)	$(1 - p + pe^t)^n$
Poisson(λ)	$e^{\lambda(e^t-1)}$
Normal(μ, σ^2)	$e^{\mu t + \frac{\sigma^2 t^2}{2}}$
Exponential(λ)	$\frac{\lambda}{\lambda-t}$ for $t < \lambda$
Gamma(α, β)	$\left(\frac{\beta}{\beta-t}\right)^\alpha$ for $t < \beta$
Chi-square(ν)	$(1 - 2t)^{-\nu/2}$ for $t < \frac{1}{2}$

A.2. A.2 Moment Generating Functions.

Distribution	Parameter	Sufficient Statistic
Normal(θ, σ^2)	θ (σ^2 known)	$\bar{X}$
Normal(μ, θ)	θ (μ known)	$\sum (X_i - \mu)^2$
Normal(θ_1, θ_2)	(θ_1, θ_2)	$(\bar{X}, S^2)$ or $(\sum X_i, \sum X_i^2)$
Poisson(θ)	θ	$\sum X_i$
Exponential(θ)	θ	$\sum X_i$
Uniform($0, \theta$)	θ	$\max(X_i)$
Bernoulli(θ)	θ	$\sum X_i$
Gamma(α, θ)	θ (α known)	$\sum X_i$

A.3. A.3 Sufficient Statistics for Common Distributions.

Distribution	$\alpha = 0.10$	$\alpha = 0.05$	$\alpha = 0.01$
Standard Normal $z_{\alpha/2}$	1.645	1.96	2.576
$t(10)$	1.812	2.228	3.169
$t(20)$	1.725	2.086	2.845
$t(30)$	1.697	2.042	2.750
$\chi^2(1)$	2.706	3.841	6.635
$\chi^2(5)$	9.236	11.071	15.086
$\chi^2(10)$	15.987	18.307	23.209
$F(1, 10)$	3.285	4.965	10.044
$F(5, 10)$	2.522	3.326	5.636

A.4. A.4 Critical Values.

Hypothesis	Test Statistic	Reject H_0 if
$H_0 : \mu = \mu_0$ (σ known)	$Z = \frac{\bar{X} - \mu_0}{\sigma/\sqrt{n}}$	$ Z > z_{\alpha/2}$
$H_0 : \mu = \mu_0$ (σ unknown)	$t = \frac{\bar{X} - \mu_0}{S/\sqrt{n}}$	$ t > t_{\alpha/2}(n-1)$
$H_0 : \sigma^2 = \sigma_0^2$	$\chi^2 = \frac{(n-1)S^2}{\sigma_0^2}$	$\chi^2 < \chi_{\alpha/2}^2(n-1)$ or $\chi^2 > \chi_{1-\alpha/2}^2(n-1)$
$H_0 : \mu_1 = \mu_2$ (equal var.)	$t = \frac{\bar{X}_1 - \bar{X}_2}{S_p \sqrt{1/n_1 + 1/n_2}}$	$ t > t_{\alpha/2}(n_1 + n_2 - 2)$
$H_0 : \sigma_1^2 = \sigma_2^2$	$F = \frac{S_1^2}{S_2^2}$	$F > F_{\alpha/2}(n_1 - 1, n_2 - 1)$

A.5. A.5 Hypothesis Testing Decision Rules.

Parameter	100(1 - α)% Confidence Interval
μ (σ known)	$\bar{X} \pm z_{\alpha/2} \frac{\sigma}{\sqrt{n}}$
μ (σ unknown)	$\bar{X} \pm t_{\alpha/2}(n-1) \frac{S}{\sqrt{n}}$
σ^2	$\left[\frac{(n-1)S^2}{\chi_{1-\alpha/2}^2(n-1)}, \frac{(n-1)S^2}{\chi_{\alpha/2}^2(n-1)} \right]$
$\mu_1 - \mu_2$ (equal var.)	$(\bar{X}_1 - \bar{X}_2) \pm t_{\alpha/2}(n_1 + n_2 - 2) S_p \sqrt{\frac{1}{n_1} + \frac{1}{n_2}}$
p (large sample)	$\hat{p} \pm z_{\alpha/2} \sqrt{\frac{\hat{p}(1-\hat{p})}{n}}$

A.6. A.6 Confidence Intervals.

A.7. **A.7 ANOVA Summary.** Test statistic: $F = \frac{MSA}{MSE} \sim F(k-1, N-k)$ under H_0

A.8. **A.8 Regression Formulas. Simple Linear Regression:** $Y_i = \beta_0 + \beta_1 x_i + \epsilon_i$

- $\hat{\beta}_1 = \frac{S_{xy}}{S_{xx}} = \frac{\sum (x_i - \bar{x})(y_i - \bar{y})}{\sum (x_i - \bar{x})^2}$
- $\hat{\beta}_0 = \bar{y} - \hat{\beta}_1 \bar{x}$
- $R^2 = \frac{SSR}{SST} = 1 - \frac{SSE}{SST}$

Source	Sum of Squares	df	Mean Square
Treatment	$SSA = \sum n_i(\bar{X}_{i.} - \bar{X}_{..})^2$	$k - 1$	$MSA = \frac{SSA}{k-1}$
Error	$SSE = \sum \sum (X_{ij} - \bar{X}_{i.})^2$	$N - k$	$MSE = \frac{SSE}{N-k}$
Total	$SST = \sum \sum (X_{ij} - \bar{X}_{..})^2$	$N - 1$	-

- $r = \frac{S_{xy}}{\sqrt{S_{xx}S_{yy}}}$ (correlation coefficient)
- Test for slope: $t = \frac{\hat{\beta}_1}{\sqrt{MSE/S_{xx}}} \sim t(n - 2)$

A.9. A.9 Maximum Likelihood Estimation Steps.

- (1) Write the likelihood function: $L(\theta) = \prod_{i=1}^n f(x_i; \theta)$
- (2) Take the log-likelihood: $\ell(\theta) = \log L(\theta) = \sum_{i=1}^n \log f(x_i; \theta)$
- (3) Differentiate: $\frac{d\ell(\theta)}{d\theta} = 0$
- (4) Solve for $\hat{\theta}_{MLE}$
- (5) Verify it's a maximum (second derivative test)

A.10. A.10 Asymptotic Results.

- **CLT:** $\frac{\sqrt{n}(\bar{X}_n - \mu)}{\sigma} \xrightarrow{d} N(0, 1)$
- **MLE Asymptotic:** $\sqrt{n}(\hat{\theta} - \theta) \xrightarrow{d} N(0, I^{-1}(\theta))$
- **Delta Method:** $\sqrt{n}(g(\hat{\theta}) - g(\theta)) \xrightarrow{d} N(0, [g'(\theta)]^2 I^{-1}(\theta))$
- **LRT:** $-2 \log \Lambda \xrightarrow{d} \chi^2(r)$ where r = difference in parameters